

Week 7 Assignment - Delta Lake MERGE Implementation

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1. Objective & Overview

This assignment is split into two parts. Part 1 covers Python and Pandas basics — loading a CSV dataset, exploring it, cleaning it, and exporting the cleaned result. Part 2 performs incremental data processing using Delta Lake: loading data into a Delta table, simulating incremental records, applying a MERGE (upsert) operation, validating the results, and implementing Slowly Changing Dimension (SCD) Type 2 to preserve historical versions of changed records.

Dataset used: **Sample - Superstore** (customer_master.csv), loaded via Databricks Volumes.

2. Steps Performed

Part 1 — Python & Pandas Basics

- Loaded customer_master.csv into a Pandas DataFrame.
- Explored the data using head(), tail(), shape, columns, and dtypes.
- Checked for missing values and handled them using fillna() (numeric columns filled with 0, Postal Code with 'Unknown').
- Filtered rows (Region = West) and selected a subset of columns for inspection.
- Removed duplicate rows using drop_duplicates().
- Created a derived column: total_amount = Sales * Quantity.
- Saved the cleaned dataset as customer_master_cleaned.csv.

Part 2 — Delta Lake MERGE (SCD Type 1)

- Loaded the master dataset into a Spark DataFrame, cleaned column names, cast data types, and re-created total_amount.
- Saved the data as a Delta table: week_7_db.customer_master.
- Created an incremental dataset simulating 20 updated records (Sales +10%) and 10 brand-new customer records (30 total).
- Applied a MERGE operation using DeltaTable.merge() — whenMatchedUpdate() for existing Row_IDs, whenNotMatchedInsertAll() for new ones.
- Validated the merge: row counts before/after, duplicate Row_ID check, and null checks on key columns.

Part 3 — SCD Type 2 (Historical Tracking)

- Added is_current, effective_date, and end_date columns to the Delta table.

- On MERGE match, expired the old row (is_current = false, end_date = current_date()) instead of overwriting it.
- Appended the incremental records as new 'current' rows, preserving full history.
- Exported the final SCD2 table as customer_master_scd2.csv.

3. Key Results & Numbers

Part 1: Pandas Cleaning Summary

Metric	Value
Original rows / columns	9,994 rows / 21 columns
Missing values found	0 (dataset was already clean)
Duplicates removed	0
West region records (filter test)	3,203
Rows after adding total_amount	9,994 rows / 22 columns
Avg / Max / Min total_amount	1,149.50 / 135,830.88 / 0.44
Output file	customer_master_cleaned.csv

Part 2: MERGE (SCD Type 1) Summary

Metric	Value
Master rows before MERGE	9,994
Incremental records applied	30 (20 updated + 10 new)
Master rows after MERGE	10,004
New rows inserted	10 (matches expected)
Duplicate Row_IDs after MERGE	0
Nulls in Sales / total_amount	300 each (from newly inserted test rows without values set).

Part 3: SCD Type 2 Summary

Metric	Value
Total rows (all versions)	10,034
Current rows (is_current = true)	10,004
Historical / expired rows	30
Avg Sales (current rows)	\$237.26
Total Revenue (current rows)	\$11,542,976.19

Top Region	West
Top Category	Office Supplies
Output file	customer_master_scd2.csv

4. Screenshot Reference

All screenshots are stored under the **screenshots/** folder, organized by stage:

Folder	Screenshot	What it shows
data_loading	01_data_loaded.png	CSV loaded into Pandas DataFrame
data_loading	02_data_explored.png	head/tail, shape, columns, dtypes output
data_loading	03_derived_column.png	total_amount column created
data_loading	04_saved_csv.png	Cleaned CSV export confirmation
data_cleaning	01_missing_values.png	Missing value check output
data_cleaning	02_filter_select.png	Filtered rows (Region = West) & selected columns
data_cleaning	03_duplicates.png	Duplicate removal before/after counts
scd1	01_master_table_created.png	Delta master table creation (week_7_db.customer_master)
scd1	02_incremental_created.png	Incremental dataset (30 records) created
scd1	03_merge_execution.png	MERGE execution + row counts before/after
scd2	scd2_implementation.png	SCD2 columns added & historical MERGE logic
validation	validation.png	Row count, duplicate, and null validation output
final_output	01_summary.png	Final assignment summary printout
final_output	02_summary_stats.png	Avg sales, total revenue, top region/category
final_output	03_final_table_category_wise.png	Final table grouped/viewed by category

5. Conclusion

The assignment successfully demonstrates the full lifecycle of incremental data processing: cleaning raw data with Pandas, loading it into a Delta Lake table, applying ACID-compliant MERGE operations to handle both updates and inserts, validating the results with row-count and integrity checks, and finally implementing SCD Type 2 to retain a complete historical record of all changes.